Shuchang Xu

Hong Kong, China sxuby@connect.ust.hk shuchangxu.github.io

Education

The Hong Kong University of Science and Technology — Hong Kong, China
Ph.D. in Computer Science, Advisor: Dr. Huamin Qu — Aug 2023 - Present

Tsinghua University — Beijing, China
M.S. in Computer Science, Advisor: Dr. Yuanchun Shi — Aug 2018 - Jun 2021

Tsinghua University — Beijing, China
B.S. in Electrical Engineering, GPA: 91.2/100 (Top 5%) — Aug 2014 - Jun 2018
B.S. in Digital Media Art and Design (Second Major)

Research Interests

As a researcher in **Human-Computer Interaction**, I focus on leveraging **Multi-Modal Machine Learning** and **Wearable Interfaces** to advance **Visual Accessibility** across both digital and physical environments.

Conference Publications

[1] **Branch Explorer: Leveraging Branching Narratives to Support Interactive 360° Video Viewing for Blind and Low Vision Users.**
Shuchang Xu, Xiaofu Jin, Wenshuo Zhang, Huamin Qu, Yukang Yan.
UIST 2025 - ACM Symposium on User Interface Software and Technology.

[2] **DanmuA11y: Making Time-Synced On-Screen Video Comments (Danmu) Accessible to Blind and Low Vision Users via Multi-Viewer Audio Discussions.**
Shuchang Xu, Xiaofu Jin, Huamin Qu, Yukang Yan.
CHI 2025 - ACM Conference on Human Factors in Computing Systems.

[3] **Memory Reviver: Supporting Photo-Collection Reminiscence for People with Visual Impairment via a Proactive Chatbot.**
Shuchang Xu, Chang Chen, Zichen Liu, Xiaofu Jin, Linping Yuan, Yukang Yan, Huamin Qu.
UIST 2024 - ACM Symposium on User Interface Software and Technology.

[4] **RhythmTA: a Visual-Aided Interactive System for ESL Rhythm Training via Dubbing Practice.**
Chang Chen, Sicheng Song, **Shuchang Xu**, Zhicheng Li, Huamin Qu, Yanna Lin.
UIST 2025 - ACM Symposium on User Interface Software and Technology.

[5] **NeuroSync: Intent-Aware Code-Based Problem Solving via Direct LLM Understanding Modification.**
Wenshuo Zhang, Leixian Shen, **Shuchang Xu**, Jindu Wang, Jian Zhao, Huamin Qu, Linping Yuan.
UIST 2025 - ACM Symposium on User Interface Software and Technology.

[6] **LightGuide: Directing Visually Impaired People along a Path Using Light Cues.**
Ciyuan Yang, **Shuchang Xu (co-first)**, Tianyu Yu, Guanhong Liu, Chun Yu, Yuanchun Shi.
IMWUT 2021 - Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies.

[7] **Virtual Paving: Rendering a Smooth Path for People with Visual Impairment through Vibrotactile and Audio Feedback.**
Shuchang Xu, Ciyuan Yang, Wenhao Ge, Chun Yu, Yuanchun Shi.
IMWUT 2020 - Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies.

[8] **Tactile Compass: Enabling Visually Impaired People to Follow a Path with Continuous Directional Feedback.**
Guanhong Liu, Tianyu Yu, Chun Yu, Haiqing Xu, **Shuchang Xu**, Ciyuan Yang, Feng Wang, Haipeng Mi, Yuanchun Shi.
CHI 2021 - ACM Conference on Human Factors in Computing Systems.

[9] **Accurate and Low-Latency Sensing of Touch Contact on Any Surface with Finger-Worn IMU Sensor.**
Yizheng Gu, Chun Yu, Zhipeng Li, Weiqi Li, **Shuchang Xu**, Xiaoying Wei, Yuanchun Shi.
UIST 2019 - ACM Symposium on User Interface Software and Technology.

[10] **A Machine Learning Framework to Identify Routing Short Violations from a Placed Netlist.**
Aysa Tabrizi, Nima Darav, **Shuchang Xu**, Logan Rakai, Ismail Bustany, Andrew Kennings, Laleh Behjat.
DAC 2018 - ACM/IEEE Design Automation Conference.

Poster/Demo Publications

[1] **MemeSpeak: Making Image Memes Blind-Accessible through Emotional Speech Augmentation.**
Gengyu Chen, Chiqian Xu, Junru Jiang, Yuting Lu, Xiao Wang, Huamin Qu, **Shuchang Xu**, Guanhong Liu.
CSCW 2025 Companion - ACM Conference on Computer-Supported Cooperative Work and Social Computing.

[2] **From Audio Description to Movement: Challenges and Design Principles in Video-based Body Exercise for Blind and Low Vision Users.**
Ziyi Jiang, Mengjie Jian, Jiajia Hu, Hongze Zhao, Huamin Qu, **Shuchang Xu**, Guanhong Liu.
CSCW 2025 Companion - ACM Conference on Computer-Supported Cooperative Work and Social Computing.

[3] **When Accessibility Becomes a Trap: A User-Centric Characterization of Dark Patterns Arising from Screen Reader Users' Perceived Deception in Mobile Interfaces.**
Dai Shi, Lin Tian, Jiaxun Sun, Tomimi Kawakami, Nuo Cheng, **Shuchang Xu**, Guanhong Liu.
CSCW 2025 Companion - ACM Conference on Computer-Supported Cooperative Work and Social Computing.

Fellowship & Awards - Ph.D.

Hong Kong Postgraduate Fellowship 2024–2027
Full stipend, approximately USD 50,000 per year

Best Paper Honorable Mention Award Apr 2025
CHI 2025, among the top 5% out of 5020 submissions

HKUST RedBird Academic Excellence Award 2024–2025
HKD 40,000 for outstanding academic and research performance

Research Travel Awards 2024–2025
HKD 14,000 for CHI 2025; HKD 14,700 for UIST 2024

Special Recognition for Outstanding Reviews
CHI 2025, IMWUT 2025, ISMAR 2025, CHI 2025 LBW

Fellowship & Awards - Master & Undergraduate

Outstanding Graduate in Computer Science, Tsinghua University Jun 2021
China National Scholarship for Postgraduates (Top 1%) Nov 2020
China National Scholarship for Postgraduates (Top 1%) Nov 2019
China National Scholarship for Undergraduates (Top 1%) Nov 2017
Xiaomi Artificial Intelligence Hackathon, 2nd Prize Jul 2022
Best Paper Award for HCI in Games, HCI International 2019 Jul 2019
Microsoft Mixed Reality Development Contest, 2nd Prize Dec 2017

Teaching Experiences

Guest Lecturer, User Experience Design, Tongji University 2025
Teaching Assistant, Java Programming Language, HKUST 2025
Teaching Assistant, User Experience Design, Tsinghua University 2020
Teaching Assistant, Big Data and Machine Intelligence, Tsinghua University 2019

Invited Talks

UIUC SALT Lab, Invited by Dr. Yun Huang Jun 2025
Improving Visual Media Accessibility with AI-Assisted Interactive Storytelling Techniques
Tongji University, Invited by Dr. Guanhong Liu Apr 2025
AI-Powered Interactive Systems for Improving Visual Media Accessibility

Professional Experiences

Xiaomi – **Software Product Manager** – Beijing, China Aug 2021 - Jun 2023
Bytedance – **TikTok Technical Artist** (Intern) – Shanghai, China Jun 2020 - Aug 2020
Tencent – **Game Level Designer** (Intern) – Shenzhen, China Jun 2019 - Aug 2019
NetEase – **Game Interaction Designer** (Intern) – Hangzhou, China Jun 2018 - Aug 2018

Patents

Vision Accessibility Techniques: CN111494175
Mobile Game Interaction Techniques: CN109316745, CN109460179, CN109550242
Smartphone Interaction Techniques: CN115857375, CN115755635

Skills

Programming: C#, Swift, Java, Python, React Native, C++, OpenGL, OpenCV

Platforms/IDE: Unity3D, Xcode, Android Studio

Prototyping: Hardware (Arduino, Rasperry PI); 3D (3ds Max, Maya), UX Design (After Effects, Figma)

TOEFL: 109/120 (R 29/30, L 26/30, S 24/30, W 30/30)